

Reversion Beyond Archimedean Valuations

Strong Hahn convergence, sharp Newton depth,
and minimal finite coefficient certificates

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Abstract

A positive valuation gain need not become cofinal in a non-Archimedean ordered group. Consequently, an argument that proves reversion by sending that gain to infinity can fail even when the inverse exists and every one of its coefficients has a finite exact computation. We resolve the fixed-shift Hahn power-logarithmic version of the ProveIt research question on reversion beyond Archimedean exponent groups.

The replacement invariant is the maximum length of a positive support word representing a prescribed exponent. The all-word form of Neumann’s lemma makes this length finite without any Archimedean hypothesis. It yields a complete support filtration, a finite-dimensional implicit-system theorem, and the ordinary integer-indexed Lagrange–Bürmann inverse. For Newton iteration we prove the sharper support-depth law $r \mapsto 2r + 1$ and the uniform bound $r_n = 2^{n+1} - 1$, with an example attaining every bound. We characterize exactly when valuation convergence holds uniformly for this class: the multiples of its least support exponent must be cofinal. In contrast, strong Hahn convergence always holds under the stated support hypotheses.

For any finite set of requested coefficients we construct a finite-rank quotient algebra that is minimal among algebra quotients retaining those coefficient maps. This gives terminating exact certificates even when a valuation cutoff contains infinitely many monomials. A rank-two infinite support has depth $n^2 + k + 1$ at (n, k) and admits no positive real-valued additive grading. More generally, every positive-integer superadditive profile is realized by such a rank-two support, allowing arbitrarily fast depth growth. Proofs, exact symbolic checks, a statement-level formalization plan, and further research questions are supplied. Generalized-series implicit functions are classical; the contribution here is the explicit repository resolution, sharp precision theory, and certificate construction, not a claim of first discovery of Hahn-series inversion.

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1 Research target, contribution, and scope

1.1 The repository question

The inspected repository snapshot is ProveIt commit `3f2d57344dc6f0b3dfa376f9490bfe13d1beb81e`, retrieved on 29 September 2026. Its transseries development has moved to `Analysis/Transseries`. The recent article *Support-Controlled Reversion and the Exact One-Exponential Substitution Group* asks, in its further-research section, what replaces the real-valued support bound and the sequential fixed-point argument when the exponent group is arbitrary. It explicitly separates failure of cofinal valuation improvement from failure of existence [1, 2].

That is the principal target here. The same article also proposes proof-producing support and precision inference. Our finite quotient construction addresses a precise algebraic part of that second target: which coefficients must be retained, why the retained set is finite, and what finite precision suffices for exact Newton inversion.

The inspection was targeted: the recent article, the transseries README, and the well-based-support Lean module were read. The much larger canonical volume was not exhaustively audited. The immutable source addresses in the bibliography and the accompanying provenance file identify the exact materials used. We make no claim to have compared every statement against every document in the repository.

Editorial note (ProveIt, 2026-09-29). Scope. “We resolve” in the abstract means: this article answers the research question “Beyond Archimedean exponent groups” of `Analysis/Transseries/docs/series-and-transseries/Support_Controlled_Reversion_One_Exponential/reversion_and_one_exponential.tex` for fixed-shift Hahn power-logarithmic derivations, as stated in this section. It is not the first non-Archimedean Hahn reversion in the repository: Lemma `e:lem-reversion` of the surcomplex analysis volume [9] (`Algebra/SurrealNumbers/docs/surcomplex/analysis/article.tex`) already proves existence, uniqueness and the same positive-integer Lagrange sum for Hahn sections over a non-Archimedean exponent group, by the finite-word form of Neumann’s lemma, without logarithmic blocks, characters or depth bounds; and the all-word lemma is checked in Lean (see the note after Lemma 2.2). *Priority of the Newton rate.* The law $r \mapsto 2r + 1$, $r_n = 2^{n+1} - 1$, called “sharper” in the abstract, is for $\Gamma \subseteq \mathbb{R}$ and $h = 1$ exactly the rate $e_{k+1} \geq 2e_k + \alpha + 2$, $e_k \geq (2^{k+1} - 1)\delta - 1$ of Theorem 4.1 (`thm:newton`, equation `eq:newtonrate`) of that reversion article, expressed in units of δ ; the article does not cite it. New here are its validity without cofinality, its attainment (Theorem 9.1), the valuation and cut criteria, the minimal quotients and the rank-two depth profiles. *Uncited canonical remarks.* Remark `plt:rmk:ext-archimedean` of the canonical volume (`Analysis/Transseries/docs/series-and-transseries/Transseries_And_Inversion/transseries_and_inversion.tex`) shows the phenomenon of Section 9 for lexicographic $\mathbb{Z}^2$ and $S = \{(0, 1)\}$, and Remark `plt:rmk:ext-neumann-general` there quotes, without proof, the all-word clause of Neumann’s lemma for non-Archimedean groups that Lemma 2.2 proves.

1.2 What is classical and what is established here

Higman’s word theorem and the associated positive-support summability principle are classical [4]. General implicit-function methods for generalized series already exist: van der Hoeven develops Noetherian operators and an implicit-function theorem, including summable recursive expansions [5]. Composition and different modes of transseries convergence are treated by Edgar [6]; a recent general Taylor expansion framework is developed by Bagayoko and Mantova [7]. The coefficient Lagrange inversion identity is classical as well [8].

Accordingly, the claim here is not that non-Archimedean formal inversion was previously unknown. We give a self-contained, quantitatively precise answer in the coefficient algebra used by the repository, with the extra data needed to define its derivative outside a real exponent

group. The central outputs are as follows.

Output	Precise content
Existence and reversion	Arbitrary ordered-group supports are allowed; ordinary integer Lagrange indices suffice; the normalized inverse stays in the positive monoid generated by the input support.
Sharp precision	Newton error depth improves from r to $2r + 1$; $r_n = 2^{n+1} - 1$ is attained by an explicit scalar inverse.
Topology boundary	Cofinality of multiples of the minimum support is necessary and sufficient for valuation convergence uniformly over the small-coefficient implicit systems considered here.
Finite certification	Each finite coefficient request factors through a minimal finite-rank quotient; a finite-support instance has a rational separating-functional certificate.
Infinite-support obstruction	A well-ordered rank-two support has exact quadratic depth and no positive real additive grading.

All these statements are proved below. The precise extent to which their combined formulation is new in the wider literature has not been established by a comprehensive priority search. The finite verification program audits examples and formulas; it is not a substitute for the proofs and is not a Lean formalization.

1.3 Four notions that must be distinguished

An inverse may exist as a Hahn series without its successive approximations converging in the full valuation topology. A finite list of coefficients may be computable even when all coefficients below a chosen valuation cutoff cannot be listed finitely. Finally, a formal Hahn series is not automatically a germ of a real or complex function.

We therefore keep four questions separate: formal existence; strong Hahn summation; valuation convergence; and analytic realization. The first three are addressed under explicit hypotheses. Borel summability, resurgence, Hardy-field realization, and a general natural derivation on a full logarithmic-exponential transseries field are not consequences claimed here.

2 Hahn series and the all-word support lemma

2.1 Conventions

Throughout, Γ is a set-sized totally ordered abelian group, written additively, and K is a field of characteristic zero. The order is translation invariant. We write $R = K[L]$, where L is an indeterminate, and

$$\mathcal{H} = R((t^\Gamma)) = \left\{ \sum_{\gamma \in T} p_\gamma(L) t^\gamma : T \subseteq \Gamma \text{ well ordered, } p_\gamma \in R \right\}.$$

Zero coefficients are omitted from the support. Our orientation is that larger exponents represent smaller monomials. For $f \neq 0$ let $v(f) = \min \text{supp } f$, and set $v(0) = +\infty$.

A family $(f_i)_{i \in I}$ is *Hahn summable* if its union of supports is well ordered and, for each exponent, only finitely many family members have a nonzero coefficient there. Its sum is defined using those finite coefficient sums. Convergence of an infinite sum of scalars does not make an otherwise inadmissible family Hahn summable.

In this article, a sequence f_n converges strongly in the Hahn sense to f if its supports and that of f lie in one well-ordered set and every coefficient is eventually exactly the corresponding coefficient of f . For such a sequence the successive-difference family is Hahn summable, and

$$f = f_0 + \sum_{n \geq 0} (f_{n+1} - f_n).$$

This is the explicitly used sequential notion; it must not be replaced by valuation convergence without proof.

Lemma 2.1 (Binary support calculus). *If $T, U \subseteq \Gamma$ are well ordered, then $T + U$ is well ordered and, for every $\alpha \in \Gamma$, there are only finitely many pairs $(\tau, u) \in T \times U$ satisfying $\tau + u = \alpha$.*

Proof. Every sequence in a well-ordered set has an infinite nondecreasing subsequence. Indeed, if some value occurs infinitely often use it; otherwise successively choose a minimum of a remaining tail and pass beyond its last occurrence. If a strictly decreasing sequence in $T + U$ existed, choose one representation of each member and pass to a nondecreasing subsequence of first coordinates. The second coordinates would strictly decrease, a contradiction.

For infinitely many different representations of a fixed α , the first coordinates are different. A nondecreasing subsequence is then strictly increasing, forcing the corresponding second coordinates to strictly decrease. This proves finiteness of the fiber. $\square$

It follows that the Cauchy product is well defined on $\mathcal{H}$. It is an integral domain, with $v(fg) = v(f) + v(g)$ for nonzero f, g . The following stronger statement concerns words of unbounded length, not merely a fixed finite product.

Lemma 2.2 (Positive all-word lemma). *Let $S \subseteq \Gamma_{>0}$ be well ordered. Its additive monoid*

$$M(S) = \{0\} \cup \bigcup_{n \geq 1} nS$$

is well ordered. Each $\alpha \in \Gamma$ is the sum of only finitely many ordered finite words over S . The empty word is permitted and has sum zero.

Proof. Order finite words by subsequence embedding with coordinatewise comparison: $(a_1, \dots, a_m)$ embeds in $(b_1, \dots, b_n)$ if there are $1 \leq j_1 < \dots < j_m \leq n$ with $a_i \leq b_{j_i}$. The word theorem says that every infinite sequence of words has an earlier word embedding in a later one. For completeness, its well-ordered-alphabet case is proved here by the minimal-bad-sequence argument.

Suppose a bad sequence exists. Choose its first word of minimal possible length among words beginning a bad sequence, then its second word of minimal length among continuations of the chosen prefix, and continue. No word is empty. Write $w_i = v_i a_i$, separating the last letter, and choose indices $i_0 < i_1 < \dots$ on which a_{i_j} is nondecreasing. The sequence

$$w_0, \dots, w_{i_0-1}, v_{i_0}, v_{i_1}, v_{i_2}, \dots$$

cannot be bad, because its first new word is shorter than the word chosen minimally at that position. An embedding among its old words contradicts badness of the original sequence. An old word embedding in v_{i_j} also embeds in w_{i_j} . Finally, an embedding $v_{i_j} \preceq v_{i_k}$ extends, using $a_{i_j} \leq a_{i_k}$, to $w_{i_j} \preceq w_{i_k}$. Each possibility is impossible. This proves the word theorem in the needed case.

Now an embedding of words makes their sums nondecreasing. Moreover, their sums are strictly increasing unless the two words are identical: an omitted letter is positive, and a strictly larger matched letter also strictly increases the sum. Therefore words representing a strictly

decreasing sequence of sums would be a bad sequence. This proves that $M(S)$ is well ordered. Infinitely many different words with one fixed sum would also be a bad sequence. This proves the finite-word assertion. $\square$

The combinatorial mechanism is the classical Higman–Neumann support principle [4]. The proof makes clear why no real-valued lower bound on word length is required.

Editorial note (ProveIt, 2026-09-29). Both assertions of Lemma 2.2 are already formalized in the repository’s Lean library for surreal Hahn series, over an arbitrary linearly ordered cancellative abelian group with no Archimedean hypothesis [10]. The finite-word assertion is `Surreal.HahnSeries.finite_words_of_sum_eq` (Algebra/SurrealNumbers/Surreal/HahnSeries/Neumann.lean: all finite lists over a positive partially well-ordered S with a given sum form a finite set, proved by Higman’s lemma) and, in monoid form, `Surreal.HahnSeries.finite_wordsWithSum` (Algebra/SurrealNumbers/Surreal/HahnSeries/NeumannWords.lean). Well-ordering of $M(S)$ is `Surreal.HahnSeries.isPWO_closure_of_pos` in the same file, and `Surreal.HahnSeries.neumann_positive` in `Neumann.lean` combines it with finiteness of nondecreasing words. These declarations were not visible to this article’s targeted inspection, which read only `Analysis/Transseries`; the “All-word support” layer of Section 14 can import them.

Corollary 2.3 (Anchored words). *For well-ordered $T \subseteq \Gamma$ and positive well-ordered S , the set $T + M(S)$ is well ordered. For each α , there are only finitely many pairs consisting of an anchor $\tau \in T$ and an ordered word over S whose sum with τ is α .*

Proof. Apply Lemma 2.1 to T and $M(S)$. There are finitely many possible pairs of anchor and word sum, and Lemma 2.2 supplies finitely many words for each such sum. $\square$

Remark 2.4 (Where the Archimedean proof breaks). If $\Gamma \subseteq \mathbb{R}$, $\delta = \min S > 0$, and a word has sum α , then its length is at most α/δ . In an arbitrary group, the inequalities $n\delta \leq \alpha$ may hold for every n . This invalidates that numerical bound, not Lemma 2.2. Replacing the lemma by the failed bound would discard valid inverses.

3 Support depth and a complete filtration

Assume from now on that S is nonempty, positive, and well ordered, and put $M = M(S)$. Define

$$H_S(\alpha) = \max\{n \in \mathbb{N} : \alpha \in nS\}, \quad \alpha \in M, \quad 0S = \{0\}. \quad (1)$$

Thus $H_S(0) = 0$. This is the *maximum* representation length, not the minimum number of generators. Maximum length is the relevant invariant when a coefficient may receive contributions from many nonlinear orders. Lemma 2.2 makes it finite.

For $r \geq 0$ define

$$S_{\geq r} = \bigcup_{n \geq r} nS, \quad \mathcal{A}_S = \{f \in \mathcal{H} : \text{supp } f \subseteq M\}, \quad I_r = \{f \in \mathcal{A}_S : \text{supp } f \subseteq S_{\geq r}\}. \quad (2)$$

Here $I_0 = \mathcal{A}_S$, and I_1 is its augmentation ideal: it consists precisely of the series with no exponent-zero block. We call I_r the support-depth filtration. We do not identify I_r with an ordinary algebraic ideal power without additional justification.

Proposition 3.1 (Complete support filtration). *The sets I_r are ideals of $\mathcal{A}_S$, and*

$$I_{r+1} \subseteq I_r, \quad I_p I_q \subseteq I_{p+q}, \quad \bigcap_{r \geq 0} I_r = \{0\}.$$

The ring $\mathcal{A}_S$ is complete for this filtration. If $f_n \in I_{r_n}$ and $r_n \rightarrow +\infty$ as integers, then (f_n) is Hahn summable. Every element of $1 + I_1$ is a unit, with inverse given by its geometric series.

Proof. Concatenating words gives $S_{\geq p} + S_{\geq q} \subseteq S_{\geq p+q}$; concatenating an arbitrary additional word shows that each I_r is an ideal. If f belongs to every I_r , its coefficient at α must vanish as soon as $r > H_S(\alpha)$, proving separatedness.

For the family (f_n) , the union of supports lies in the well-ordered set M . A fixed α can occur only for indices with $r_n \leq H_S(\alpha)$, of which there are finitely many. Hence the family is Hahn summable.

For completeness, let (g_n) be filtration-Cauchy. For every α , the coefficients of g_n at α eventually stabilize: use any filtration index greater than $H_S(\alpha)$. Let g consist of these stabilized coefficients. Its support is a subset of M , so it lies in $\mathcal{A}_S$. Fix r . Beyond a Cauchy index, $g_n - g_m \in I_r$ for all n, m . Passing coefficientwise to the stabilized values gives $g_n - g \in I_r$. This proves completeness.

Finally, if $a \in I_1$, then $a^n \in I_n$, so $\sum_{n \geq 0} (-a)^n$ is summable. The usual finite geometric identity, followed coefficientwise, shows that this is $(1 + a)^{-1}$. $\square$

The same conclusions hold entrywise for finite vectors and matrices. In particular, $\text{Id} + J$ is invertible whenever every entry of J is in I_1 .

Lemma 3.2 (Coefficient stopping rule). *If $f - g \in I_r$ and $r > H_S(\alpha)$, then $[t^\alpha] f = [t^\alpha] g$. For a finite nonempty target set $E \subseteq M$, it suffices that*

$$r > H_E, \quad H_E = \max_{\alpha \in E} H_S(\alpha).$$

Proof. No exponent of depth less than r is in $S_{\geq r}$. $\square$

Notice that this is a coefficient certificate, not a claim that every initial segment of M is finite. The distinction persists even when S itself is finite.

4 Small-coefficient implicit systems

The support mechanism is not restricted to one inverse equation. A finite system makes the source of the extra Newton gain particularly transparent.

Fix $d \geq 1$ and formal series in d unknowns

$$A_i(U) = \sum_{\nu \in \mathbb{N}^d} a_{i,\nu} U^\nu, \quad \text{supp } a_{i,\nu} \subseteq S, \quad 1 \leq i \leq d. \quad (3)$$

There need not be a uniform bound on the degrees of the coefficient polynomials in L . The coefficients need not be summable as ν varies before evaluation. Their common positive support bound is the hypothesis. Set

$$\Psi(U) = U + A(U).$$

Lemma 4.1 (Evaluation and nonlinear estimates). *For $U \in I_1^d$, the expressions $A(U)$ and all their formal partial derivatives are defined by Hahn-summable families. Their values and all entries of $A'(U)$ lie in I_1 . For $U, V \in I_1^d$ with $U - V \in I_r^d$, $r \geq 1$, one has*

$$A(U) - A(V) \in I_{r+1}^d, \quad (4)$$

$$A(U) - A(V) - A'(V)(U - V) \in I_{2r+1}^d. \quad (5)$$

Proof. A term with $|\nu| = m$ in $A(U)$ belongs to I_{m+1} . There are finitely many multi-indices of each total degree because d is finite. The common support is contained in M , and a fixed exponent permits only finitely many such degrees by its finite depth. Each fixed-degree expression is an ordinary finite sum of Hahn products. This proves summability. The same argument applies

after any fixed number of partial differentiations; the coefficient still supplies at least one support letter.

For each monomial, $U^\nu - V^\nu$ is a sum of terms with at least one factor from $U - V$, and the associated coefficient supplies a letter from S . This proves (4). Subtracting the linear part leaves at least two factors from $U - V$, so its support depth is at least $2r + 1$. All expansions are legitimate by the preceding fixed-coefficient finiteness argument. $\square$

Theorem 4.2 (Implicit solution and sharp Newton estimate). *Under (3), the equation $\Psi(U) = 0$ has a unique solution $U_* \in I_1^d$. Picard iteration*

$$P_0 = 0, \quad P_{n+1} = -A(P_n)$$

converges strongly in the Hahn sense to U_ , with*

$$P_n - U_* \in I_{n+1}^d. \quad (6)$$

Newton iteration

$$N_0 = 0, \quad N_{n+1} = N_n - [\text{Id} + A'(N_n)]^{-1} \Psi(N_n) \quad (7)$$

is well defined and converges strongly to the same solution. Its error satisfies

$$N_n - U_* \in I_{r_n}^d, \quad r_n = 2^{n+1} - 1. \quad (8)$$

More generally, one Newton step sends an error in I_r^d into I_{2r+1}^d .

Proof. Picard iteration stays in I_1^d . Its first increment belongs to I_1^d , and (4) shows inductively that $P_{n+1} - P_n \in I_{n+1}^d$. Proposition 3.1 gives a strong limit $U_* \in I_1^d$. The same Lipschitz estimate permits passage to the limit in $P_{n+1} = -A(P_n)$, yielding $\Psi(U_*) = 0$. The tail of the increment family gives (6).

If $U, V \in I_1^d$ are solutions, their difference first belongs to I_1^d . Equation (4) and the fixed-point equation then put it in I_2^d , then in I_3^d , and so forth. Separatedness makes it zero. No valuation cofinality enters this argument.

For Newton iteration, $J(U) = \text{Id} + A'(U)$ is a matrix over $\mathcal{A}_S$ congruent to Id modulo I_1 , hence is invertible by its matrix geometric series. The correction belongs to I_1^d , so the iteration stays in that domain. Suppose $e = U - U_* \in I_r^d$. Taylor expansion about U , evaluated at U_* , gives

$$0 = \Psi(U_*) = \Psi(U) - J(U)e + R, \quad R \in I_{2r+1}^d.$$

The linear part U of Ψ has no quadratic remainder; all higher terms come from A , whose coefficients carry the additional support letter. Consequently,

$$U - J(U)^{-1} \Psi(U) - U_* = J(U)^{-1} R \in I_{2r+1}^d.$$

Initially $N_0 - U_* \in I_1^d$. The recurrence $r_{n+1} = 2r_n + 1$, $r_0 = 1$, has solution $r_n = 2^{n+1} - 1$. The resulting errors and increments form Hahn-summable families, and each coefficient eventually agrees with U_* . This proves strong convergence. $\square$

Editorial note (ProveIt, 2026-09-29). Applied to reversion (Section 9), where the normalized correction is $U = t^h C$, with $\Gamma \subseteq \mathbb{R}$ and $h = 1$, (8) gives $v(C_n - C_*) \geq (2^{n+1} - 1)\delta - 1$, $\delta = \min S$, and the step $r \mapsto 2r + 1$ becomes $e \mapsto 2e + \alpha + 2$ with $\delta = \alpha + 1$: this is equation `eq:newtonrate` of Theorem 4.1 (`thm:newton`) in `Analysis/Transseries/docs/series-and-transseries/Ssupport_Controlled_Reversion_One_Exponential/reversion_and_one_exponential.tex`, which this article does not cite. The depth formulation extends that rate to groups in which the multiples of δ are not cofinal; Theorem 9.1 shows it cannot be improved.

Corollary 4.3 (Exact Newton round count). *For a finite nonempty set of requested exponents $E \subseteq M$, all those coefficients of N_n are exact whenever*

$$2^{n+1} - 1 > H_E. \quad (9)$$

In particular,

$$n = \lfloor \log_2(H_E + 1) \rfloor$$

is sufficient, including $n = 0$ when $H_E = 0$.

Proof. Combine (8) with Lemma 3.2. The displayed integer is the least nonnegative n satisfying (9). $\square$

Proposition 4.4 (Uniqueness beyond the generated monoid). *The solution in Theorem 4.2 is the only solution whose components are arbitrary positive-support Hahn series, even if their supports were not initially assumed to lie in $M(S)$.*

Proof. For any finite collection of positive-support inputs, their combined support is well ordered and positive. The anchored-word lemma justifies evaluation of (3) on these inputs. Write $\delta = \min S$. If U, V are such inputs, and $\epsilon = \min_i v(U_i - V_i)$ is finite, the monomial difference factorization gives

$$v(A_i(U) - A_i(V)) \geq \epsilon + \delta \quad \text{for every } i.$$

All other factors have nonnegative valuations, and every coefficient of A has valuation at least δ . If both inputs were solutions, some component of $U - V$ would have valuation ϵ , whereas its equal negative feedback difference would have strictly larger valuation. This is impossible. A *single* strict gain suffices for uniqueness; its multiples need not be cofinal. $\square$

5 An explicit finite-tree coefficient formula

The fixed-point proof has a useful nonrecursive interpretation. Write

$$a_{i,\nu} = \sum_{s \in S} a_{i,\nu,s}(L)t^s.$$

Consider finite rooted trees whose vertices have types in $\{1, \dots, d\}$. A vertex of type i has ν_j ordered child slots of type j , grouped in increasing order of j , for some $\nu \in \mathbb{N}^d$. Give this vertex an additional label $s \in S$ and weight $-a_{i,\nu,s}(L)$. Zero-weight decorations may simply be omitted. The exponent of a decorated tree is the sum of its vertex labels; its coefficient weight is the product of its vertex weights.

Theorem 5.1 (Finite tree formula). *For $\alpha \in M$ and root type i ,*

$$[t^\alpha] U_{*,i} = \sum_{\substack{\mathcal{T} \text{ decorated tree of root type } i \\ \sum_{v \in \mathcal{T}} s_v = \alpha}} \prod_{v \in \mathcal{T}} (-a_{i_v, \nu_v, s_v}(L)). \quad (10)$$

This is a finite sum, and every contributing tree has at most $H_S(\alpha)$ vertices. There is no automorphism-group denominator: child slots are ordered as specified above.

Proof. List the vertices in depth-first order. Their labels form an ordered word over S with sum α . There are finitely many such words and a finite upper bound on their lengths. For a fixed number of vertices there are finitely many rooted ordered tree shapes, finitely many vertex types, and finitely many possible child-count vectors. Thus the displayed family is finite at each exponent and has support in M .

Decompose a tree at its root. The root supplies one coefficient of $-A_i$; the ordered subtrees in its child slots supply the factors $U_1^{\nu_1} \cdots U_d^{\nu_d}$. This is exactly the fixed-point equation. Conversely, expanding a product in that equation independently chooses a subtree for each slot, producing every decorated tree with precisely its multiplicity. Hence the Hahn sum over all trees is a solution in I_1^d . Uniqueness proves (10). $\square$

The tree formula is not presented as a new general theory of implicit functions. Its role here is to exhibit a finite certificate for an individual coefficient: the dependence on arbitrarily high nonlinear orders disappears once the target exponent is fixed. Picard iteration organizes the same trees by height. The support-depth bound instead controls their number of vertices, and remains useful when infinitely many support exponents lie below a valuation cutoff.

6 The fixed-shift power–logarithmic differential algebra

6.1 The necessary additional data

In a real exponent group, the exponent γ can also be used as a scalar coefficient. This is not meaningful in an arbitrary ordered abelian group without a specified map. We therefore fix

$$h \in \Gamma_{>0}, \quad \chi : \Gamma \longrightarrow (K, +) \text{ additive}, \quad \chi(h) = 1. \quad (11)$$

No order compatibility of χ is assumed. Put $X = t^{-h}$, and define

$$D(t^\gamma p(L)) = t^{\gamma+h} (p'(L) - \chi(\gamma)p(L)). \quad (12)$$

Then $DX = 1$ and $DL = t^h$. When $\Gamma \subseteq \mathbb{R}$, $K = \mathbb{R}$, $h = 1$ and $\chi(\gamma) = \gamma$, this is exactly the power–logarithmic derivative used by the repository source.

The data (11) can always be supplied after a suitable choice of scalar map: an ordered abelian group is torsion free, hence embeds in $\Gamma \otimes_{\mathbb{Z}} \mathbb{Q}$; extend h to a rational vector-space basis and send it to 1. Sending the other basis elements to zero gives a map into $\mathbb{Q}$. If an injective additive map is desired, send them instead to algebraically independent indeterminates and take a corresponding rational-function field. These choices need not be canonical and need not provide an analytic interpretation.

Proposition 6.1 (Derivative and Euler operator). *Equation (12) extends to a strongly summation-preserving derivation of $\mathcal{H}$. For every $j \geq 0$,*

$$D^j(t^\gamma p(L)) = t^{\gamma+jh} \prod_{i=0}^{j-1} (\partial_L - \chi(\gamma) - i)p(L). \quad (13)$$

In particular $v(D^j f) \geq v(f) + jh$. The Euler operator

$$\nabla = XD = t^{-h}D \quad (14)$$

is a derivation preserving each support exponent, with $\nabla(t^\gamma p) = t^\gamma(p' - \chi(\gamma)p)$.

Proof. Translation by h preserves well ordering and finite support fibers, so termwise application is strong. The product rule follows on two monomial blocks from additivity of χ and the ordinary polynomial product rule; strong summation extends it to all series. Induction on j , using $\chi(\gamma + jh) = \chi(\gamma) + j$, gives (13). Multiplication of a derivation by an element of a commutative ring is again a derivation; multiplying (12) by t^{-h} gives the last formula. $\square$

Remark 6.2 (The scope is deliberate). The ordered exponent group controls support; the character χ controls the differential weights. A character with kernel gives additional constant monomials. Nothing in the reversion proof requires the constant field to equal K . A full transseries field can have a more complicated derivative whose support shifts vary with the monomial. Such a derivative is outside the fixed-shift theorem and is a further research direction, not silently covered by (12).

7 Strong Taylor substitution without cofinality

Let $B \in \mathcal{H}$ satisfy $v(B) > -h$, and put

$$b = t^h B, \quad H_B = X + B.$$

The support $S = \text{supp } b$ is positive and well ordered. The zero case $B = 0$ is understood separately as the identity substitution. Define

$$T_B f = f \circ H_B := \sum_{j \geq 0} \frac{B^j}{j!} D^j f. \quad (15)$$

Here the notation $\circ$ denotes the substitution being constructed, not an assumed analytic composition operation.

Theorem 7.1 (Strong substitution calculus). *For every $f \in \mathcal{H}$, formula (15) is Hahn summable. If $T = \text{supp } f$, its support is contained in $T + M(S)$. The map T_B is a strongly summation-preserving K -algebra homomorphism. It acts by*

$$T_B(t^\gamma) = t^\gamma(1+b)^{-\chi(\gamma)}, \quad T_B(L) = L + \log(1+b). \quad (16)$$

It preserves the leading coefficient block and valuation of every nonzero series. Moreover,

$$D(T_B f) = (1 + DB)T_B(Df), \quad (17)$$

$$T_C(T_B f) = T_{C+T_C B} f \quad (18)$$

whenever $v(C) > -h$.

Proof. By (13), the j th Taylor summand has support in

$$j(\text{supp } B) + (T + jh) = T + jS.$$

The anchored-word lemma proves that the union is well ordered and that a fixed target receives contributions from only finitely many Taylor orders and finitely many input anchors. This proves summability. For an arbitrary Hahn-summable input family, its combined anchors form a well-ordered set and each anchor occurs in only finitely many members. The same argument therefore permits both orders of summation and proves strong preservation.

For t^γ , the iterated derivative formula gives the binomial series in (16). For L , its positive derivatives are

$$D^j L = (-1)^{j-1} (j-1)! t^{jh} \quad (j \geq 1),$$

which give the logarithm. The binomial and logarithmic series are themselves summable because b has positive support. On monomial blocks, (16) respects products, by additivity of χ and the identity $(1+b)^{a+c} = (1+b)^a(1+b)^c$. It respects polynomial operations in L as well. These formal identities may be checked in one ordinary power-series variable and specialized using the word lemma. Strong summation then proves the homomorphism assertion on $\mathcal{H}$.

Every positive Taylor order shifts an anchor by a positive support word. It therefore contributes nothing to the least exponent of f . The leading block is unchanged.

For the chain rule, differentiating the summable Taylor family gives

$$\begin{aligned} D(T_B f) &= \sum_{j \geq 0} \frac{B^j}{j!} D^{j+1} f + (DB) \sum_{j \geq 1} \frac{B^{j-1}}{(j-1)!} D^j f \\ &= (1 + DB) T_B(Df). \end{aligned}$$

The derivative is strong and products of the relevant families are summable, justifying the displayed manipulation.

For associativity, let $c = t^h C$ and set $E = C + T_C B$. Since T_C preserves valuation, $v(E) > -h$. Also

$$t^h E = c + (1 + c) T_C b, \quad 1 + t^h E = (1 + c)(1 + T_C b).$$

Applying T_C to (16), and using this product identity, shows that $T_C T_B$ and T_E agree on every t^γ and on L . For L use $\log(uv) = \log u + \log v$ for commuting elements of $1 + I_1$ in the monoid generated by the combined positive supports. Both sides preserve Hahn sums, so they agree on every series. This proves (18). $\square$

Proposition 7.2 (A difference estimate requiring only one gain). *If $v(B) > -h$ and $v(C), v(E) > -h$, then*

$$v(T_C B - T_E B) \geq v(C - E) + v(B) + h. \quad (19)$$

In particular, $H_B \circ H_C = H_B \circ H_E$ implies $C = E$.

Proof. Subtract Taylor expansions and use $C^j - E^j = (C - E) \sum_{a=0}^{j-1} C^a E^{j-1-a}$ for $j \geq 1$. Each resulting term has valuation at least

$$v(C - E) + v(B) + h + (j - 1)(h + \min\{v(C), v(E)\}).$$

The last summand is nonnegative. All families are summable by applying the word lemma to the union of the normalized input supports. This proves (19). If $C - E = -(T_C B - T_E B)$ and $C \neq E$, the right side would have strictly larger valuation than the left, because $v(B) + h > 0$. $\square$

8 Reversion and the ordinary Lagrange index

8.1 Reduction to a small-coefficient equation

Write the normalized input as

$$b = t^h B = \sum_{s \in S} b_s(L) t^s. \quad (20)$$

Seek $G = X + C$ and put $U = t^h C$. By (16), the equation $H_B \circ G = X$ is equivalent to

$$\Psi_B(U) = U + A_B(U) = 0, \quad (21)$$

where

$$A_B(U) = \sum_{s \in S} t^s (1 + U)^{1 - \chi(s)} b_s(L + \log(1 + U)). \quad (22)$$

Indeed, a block of B has exponent $s - h$ and hence differential weight $\chi(s) - 1$. The formal coefficient of every power of U in A_B has support contained in S . Thus (22) is a scalar instance of (3).

Theorem 8.1 (Non-Archimedean power-logarithmic reversion). *Assume (11). Every $H_B = X + B$ with $v(B) > -h$ has a unique two-sided inverse of the form $G = X + C$, $v(C) > -h$. Its normalized correction $U = t^h C$ belongs to I_1 for $S = \text{supp}(t^h B)$. In particular, its support is contained in $M(S) \setminus \{0\}$.*

The inverse is given by the ordinary positive-integer Lagrange sum

$$G = X + \sum_{n \geq 1} \frac{(-1)^n}{n!} D^{n-1}(B^n). \quad (23)$$

The normalized n th summand has support in nS . Consequently, the coefficient at exponent α of U uses at most the outer orders $1 \leq n \leq H_S(\alpha)$, whether or not $n \min S$ is cofinal in Γ .

Proof. Theorem 4.2, applied to (22), gives a solution $U \in I_1$, hence a right substitution inverse $G = X + t^{-h}U$ with $H_B \circ G = X$. It lies in the same near-identity class. Apply the same existence argument to G to obtain H with $G \circ H = X$. Associativity from Theorem 7.1 gives

$$H_B = H_B \circ (G \circ H) = (H_B \circ G) \circ H = H.$$

Therefore $G \circ H_B = X$, proving two-sided inversion. Proposition 7.2 proves uniqueness among all near-identity corrections, not only those with a prespecified generated support.

To derive (23), introduce an independent ordinary formal variable z and solve

$$C(z) = -z \Phi(C(z)), \quad \Phi(y) = \sum_{j \geq 0} \frac{D^j B}{j!} y^j \quad \text{in } \mathcal{H}[[y]].$$

Classical coefficient Lagrange inversion, over any commutative $\mathbb{Q}$ -algebra, gives

$$[z^n]C(z) = \frac{(-1)^n}{n} [y^{n-1}] \Phi(y)^n.$$

A proof of the exact form used here is included in Appendix A. The formal Taylor map $f \mapsto \sum_j D^j f y^j / j!$ is a ring homomorphism, by the iterated Leibniz formula. Hence

$$[y^{n-1}] \Phi(y)^n = \frac{1}{(n-1)!} D^{n-1}(B^n),$$

which proves the coefficient formula with z retained.

It remains essential to justify setting $z = 1$. The normalized n th coefficient has support in nS : B^n shifts nS by $-nh$, D^{n-1} shifts it by $(n-1)h$, and multiplication by t^h removes the remaining shift. Lemma 2.2 therefore makes the normalized coefficient family Hahn summable. The parameter equation can be written

$$U(z) + z A_B(U(z)) = 0, \quad U(z) = t^h C(z).$$

Upon fully expanding a coefficient of this equation, each occurrence of z is accompanied by one input support letter. For any target exponent there are only finitely many decorated words, so substitution $z = 1$ commutes with every sum and product in the equation. The resulting Hahn sum solves (21); uniqueness identifies it with U . Finally, no normalized n th term can contribute at α for $n > H_S(\alpha)$. $\square$

Editorial note (ProveIt, 2026-09-29). At $\Gamma \subseteq \mathbb{R}$, $h = 1$, $\chi = \text{Id}$, (23) is Theorem 3.2 (`thm:LB`, equation `eq:LB`) of the predecessor reversion article (`Analysis/Transseries/docs/series-and-transseries/Support_Controlled_Reversion_One_Exponential/reversion_and_one_exponential.tex`), whose A and B are the B and C here, and Corollary 8.2 agrees term by term with its block formula `eq:blockLB`. Over a non-Archimedean exponent group, existence, uniqueness

and the same sum without logarithmic blocks are Lemma `e:lem-reversion` of the surcomplex analysis volume (`Algebra/SurrealNumbers/docs/surcomplex/analysis/article.tex`), proved by the same auxiliary-parameter specialization. What this theorem adds is the power-logarithmic block calculus over a non-Archimedean group with fixed shift data (h, χ) ; its outer-order bound $n \leq H_S(\alpha)$ replaces the real-valued bound `eq:rbound` of the predecessor.

8.2 A normalized Euler formula

The derivative formula can be kept entirely inside the generated support algebra, an advantage for finite quotient computation.

Corollary 8.2 (Euler-operator reversion). *With $b = t^h B$ and $\nabla = XD$, the normalized inverse is*

$$U = \sum_{n \geq 1} \frac{(-1)^n}{n!} \left(\prod_{j=2}^n (\nabla + j) \right) b^n. \quad (24)$$

The empty product at $n = 1$ is the identity. Each factor preserves support, and the n th term is supported in nS .

Proof. Since $B^n = t^{-nh} b^n$, repeated use of $D(t^\gamma f) = t^{\gamma+h}(\nabla f - \chi(\gamma)f)$ gives

$$t^h D^{n-1}(B^n) = (\nabla + 2)(\nabla + 3) \cdots (\nabla + n) b^n.$$

The factors commute because each is a polynomial in the same operator. Insert this identity in (23). $\square$

For example, the first three normalized orders are

$$U = -b + \frac{1}{2}(\nabla + 2)b^2 - \frac{1}{6}(\nabla + 2)(\nabla + 3)b^3 + \cdots.$$

These are nonlinear support orders. With resonances, several such orders can contribute to the same exponent; the formula neither chooses a free multi-index representation nor ignores these identifications.

8.3 Support and exponent-lattice stability

Corollary 8.3 (No new exponent lattice). *If S lies in a subgroup $\Lambda \subseteq \Gamma$, then the normalized inverse support lies in Λ . Every normalized Picard and Newton iterate also has support in $M(S)$. If $h \in \Lambda$, the unnormalized corrections remain in Λ as well.*

Proof. The monoid $M(S)$ is contained in Λ , and every construction takes place in $\mathcal{A}_S$. Multiplication by t^{-h} shifts exponents by $-h$. $\square$

Thus the denominator-stability conclusion in a fixed Puiseux lattice extends to arbitrary ordered-group sublattices with the stated shift qualification. This is a support assertion. It supplies neither an algorithm for an unrepresented group nor an analytic realization of its monomials.

9 Sharp Newton depth and the exact valuation boundary

9.1 The Newton iteration attached to reversion

Differentiating (22) with respect to U gives

$$A'_B(U) = \sum_{s \in S} t^s (1 + U)^{-\chi(s)} [b'_s(L + \log(1 + U)) + (1 - \chi(s))b_s(L + \log(1 + U))]. \quad (25)$$

Equivalently, if $C = t^{-h}U$, then $A'_B(U) = T_C(DB)$. Thus the Newton iteration of Theorem 4.2 is exactly

$$C_0 = 0, \quad C_{n+1} = C_n - \frac{C_n + T_{C_n}B}{1 + T_{C_n}(DB)}. \quad (26)$$

The denominator is $1 +$ a positive-support series, so it is a unit. The normalized error $t^h(C_n - C)$ lies in $I_{2^{n+1}-1}$.

9.2 A scalar model attaining every support bound

Theorem 9.1 (Sharpness at every Newton stage). *Let z be an ordinary formal variable and consider*

$$\Psi(u) = u + \frac{z}{1+u} = 0. \quad (27)$$

Its zero-constant solution is

$$u_* = \frac{-1 + \sqrt{1-4z}}{2} = - \sum_{m \geq 1} \frac{1}{m} \binom{2m-2}{m-1} z^m. \quad (28)$$

Newton iteration for the rational function (27), starting at $u_0 = 0$, satisfies

$$u_n - u_* = z^{r_n} + \mathcal{O}(z^{r_n+1}), \quad r_n = 2^{n+1} - 1. \quad (29)$$

Consequently, the uniform support-depth bound in Theorem 4.2 cannot be increased by even one at any stage.

Proof. The root formula follows from $u_*^2 + u_* + z = 0$ and the branch with constant term zero. Its coefficient expansion is obtained by the formal binomial series.

Write $e = u - u_*$. A direct algebraic simplification of the Newton update, using $z = -u_*(1+u_*)$, gives the exact error identity

$$e_{\text{new}} = \frac{-u_*e^2}{(1+u)^2 - z}. \quad (30)$$

The denominator has constant term 1, and $-u_* = z + \mathcal{O}(z^2)$. Thus an error $e = z^r + \mathcal{O}(z^{r+1})$ becomes $z^{2r+1} + \mathcal{O}(z^{2r+2})$, with leading coefficient still 1. Since $e_0 = -u_* = z + \mathcal{O}(z^2)$, induction proves (29). Take the support set to consist of the single exponent of z ; its depth is ordinary power degree. Every asserted bound is then attained. $\square$

Remark 9.2 (Do not change the equation before comparing Newton rates). The rational equation $u + z/(1+u) = 0$ and the polynomial equation $u^2 + u + z = 0$ have the same small root but different Newton maps. Multiplying an equation by a nonconstant unit need not preserve Newton iterates. The support gain $2r+1$ and the verification program concern (27), which is the equation arising from normalized reversion.

9.3 A complete criterion for uniform valuation convergence

The valuation topology on $\mathcal{H}$ has zero-neighborhoods $\{f : v(f) > \beta\}$ for $\beta \in \Gamma$. For finite vectors use the minimum component valuation. Say that $\delta > 0$ is *cofinal by multiples* if, for every $\beta \in \Gamma$, some $n \in \mathbb{N}$ satisfies $n\delta > \beta$.

Theorem 9.3 (Exact uniform valuation criterion). *Fix a nonempty positive well-ordered support set S , and let $\delta = \min S$. The following are equivalent.*

- (i) *The multiples of δ are cofinal in Γ .*

- (ii) *For every finite-dimensional small-coefficient system (3) with coefficient supports contained in S , its Newton sequence starting at zero converges in the full valuation topology.*
- (iii) *For every such system, its Picard sequence starting at zero converges in the full valuation topology.*

Strong Hahn convergence holds for every one of these systems even when these equivalent conditions fail.

Proof. Every exponent in $S_{\geq r}$ is at least $r\delta$. Therefore (6) and (8) imply valuation convergence when $n\delta$ is cofinal. This proves (i) $\Rightarrow$ (ii),(iii).

Conversely, put $z = t^\delta$ and use the scalar system $u + z/(1 + u) = 0$. Its feedback power-series coefficients are $(-1)^j z$, all supported on $\{\delta\} \subseteq S$, so it is in the stated class. By Theorem 9.1, its Newton error has valuation $r_n\delta$. If the multiples of δ are not cofinal, choose β bounding them above. None of these errors has valuation greater than β ; hence the Newton sequence does not converge in the valuation topology.

For Picard iteration $p_{n+1} = -z/(1 + p_n)$, subtraction of the fixed-point equation gives

$$p_{n+1} - u_* = \frac{z(p_n - u_*)}{(1 + p_n)(1 + u_*)}.$$

Starting from $p_0 - u_* = z + \mathcal{O}(z^2)$, this implies $p_n - u_* = z^{n+1} + \mathcal{O}(z^{n+2})$. The same bound β prevents valuation convergence. Thus (ii) and (iii) each imply (i). Strong convergence follows from Theorem 4.2 without any cofinality assumption. $\square$

The quantifier “for every system” is essential. A particular equation can have a finite exact iteration even if δ is noncofinal; for example, $u + a = 0$ is solved by one Newton update. The theorem classifies the uniform guarantee supplied by the support class, not the behavior of every individual equation separately.

9.4 An actual reversion that is not valuation-Cauchy

Take $\Gamma = \mathbb{Z}^2$ in lexicographic order and set

$$h = (0, 1), \quad \eta = (1, 0).$$

Then $nh < \eta$ for every integer n . One may take $K = \mathbb{Q}(\omega)$ with ω an indeterminate and $\chi(a, b) = a\omega + b$, so the character can be injective; the following example does not depend on a large kernel of χ .

Let

$$F = X + X^{-1}, \quad X = t^{-h}, \quad z = t^{2h}. \quad (31)$$

Its normalized inverse equation is exactly (27), with $G = X(1 + u_*)$. Hence

$$G = \frac{X + \sqrt{X^2 - 4}}{2}, \quad (32)$$

where the square root means $X\sqrt{1 - 4X^{-2}}$ in the near-identity branch. The Newton corrections satisfy

$$v(C_n - C_*) = (2r_n - 1)h = (2^{n+2} - 3)h < \eta. \quad (33)$$

They are not even valuation-Cauchy: the leading exponent of $C_{n+1} - C_n$ is the leading exponent of $C_n - C_*$, since the error at the next stage has strictly larger exponent. Nevertheless, they converge strongly to the exact inverse, and any specified coefficient becomes exact after finitely many stages.

The formal closed expression (32) is perfectly finite. What fails is a particular topological convergence requirement and the possibility of listing the infinitely many nonzero terms below the cutoff η in a finite monomial list. These are not failures of the inverse.

10 When is an entire valuation cutoff finite?

The support criterion for a finite coefficient request is automatic, but that for a whole valuation cutoff is not.

Theorem 10.1 (Exact cut-finiteness criterion). *Let $S \subseteq \Gamma_{>0}$ be nonempty and well ordered, let $\delta = \min S$, and let $\beta \geq 0$. Then*

$$M(S) \cap (-\infty, \beta] \text{ is finite} \quad (34)$$

if and only if both conditions hold:

$$S \cap (-\infty, \beta] \text{ is finite,} \quad N\delta > \beta \text{ for some } N \geq 1.$$

Proof. If (34) holds, its subset of generators is finite. Also, the distinct elements $n\delta$ belong to $M(S)$. If every $n\delta \leq \beta$, the cutoff would contain infinitely many of them, a contradiction.

Conversely, suppose the two conditions hold. A word of sum at most β has length less than N , because each letter is at least δ . Every letter of such a word is at most β , because all letters are positive. There are only finitely many permitted letters and finitely many permitted lengths, hence only finitely many words and sums. $\square$

For $\beta < 0$ the intersection is empty. For finite S , only the cofinality-across-that-cut condition remains. In the rank-two example, $S = \{2h\}$ is finite but no multiple of $2h$ crosses η , so the cutoff is infinite.

Remark 10.2 (Existential finiteness is not an input representation). For an arbitrary infinite support supplied with no effective description, Lemma 2.2 proves that a coefficient has finitely many contributors; it does not provide an algorithm to discover them. Effective computation requires a presentation of the support and exponent arithmetic, or a finite certificate that its relevant part is complete. The next section supplies such certificates for finite generator lists, and Section 12 gives an explicit infinite-support family.

11 Minimal finite coefficient quotients and certificates

11.1 The exact dependence set

Let $E \subseteq M(S)$ be finite and nonempty. Define its monoid-divisor closure by

$$D_S(E) = \{\beta \in M(S) : \alpha - \beta \in M(S) \text{ for some } \alpha \in E\}. \quad (35)$$

The notation involves group subtraction, but both summands in a permitted decomposition must lie in the positive monoid. This is not an order interval: a smaller exponent need not be a monoid divisor.

Lemma 11.1 (Finite dependence closure). *The set $D_S(E)$ is finite, contains 0 and E , and is downward closed under monoid divisibility. Every $\beta \in D_S(E)$ satisfies $H_S(\beta) \leq H_E$.*

Proof. For each target α , Lemma 2.1 gives finitely many pairs $(\beta, \gamma) \in M \times M$ with $\beta + \gamma = \alpha$. Taking the finite union over E proves finiteness. The containment assertions follow by using zero

as a summand. If β divides α and β' divides β , then β' divides α , proving downward closure. A maximum-length word for β can be concatenated with a word for $\alpha - \beta$; its length is at most $H_S(\alpha) \leq H_E$. $\square$

Put $D_E = D_S(E)$ and define

$$J_E = \{f \in \mathcal{A}_S : [t^\beta] f = 0 \text{ for all } \beta \in D_E\}. \quad (36)$$

Theorem 11.2 (Minimal coefficient-retaining quotient). *The set J_E is an ideal, and*

$$Q_E = \mathcal{A}_S / J_E \cong \bigoplus_{\beta \in D_E} R t^\beta \quad (37)$$

as R -modules. Multiplication is ordinary convolution followed by restriction to D_E . This quotient is the coarsest algebra quotient that retains the coefficient maps at every target in E : if an ideal J has

$$[t^\alpha] f = 0 \quad \text{for every } f \in J \text{ and } \alpha \in E,$$

then $J \subseteq J_E$. Thus every algebra quotient through which those coefficient maps descend surjects onto Q_E .

The image of I_1 in Q_E is nilpotent of exponent at most $H_E + 1$, and the image of I_{H_E+1} is zero. The Euler derivation ∇ descends to Q_E .

Proof. If $\beta \notin D_E$ and $\gamma \in M$, then $\beta + \gamma \notin D_E$: otherwise adding its complement to a target would also make β a divisor of that target. The complement of D_E is therefore closed under addition of M . It follows that J_E is an ideal. Restriction to D_E identifies every coset with a unique finite sum of the displayed basis monomials, proving (37) and the multiplication formula. Associativity is inherited from the quotient; discarding exponents outside D_E cannot remove a contribution that later returns to a retained exponent.

For minimality, take $f \in J$ and any $\beta \in D_E$. Choose $\alpha \in E$ and $\gamma \in M$ with $\beta + \gamma = \alpha$. Since J is an ideal, $t^\gamma f \in J$. Its coefficient at α is exactly the coefficient of f at β , and must be zero. Therefore $f \in J_E$. This proves the universal quotient property for all such ideals, not only monomial ideals.

Every exponent in D_E has depth at most H_E by Lemma 11.1. Hence I_{H_E+1} maps to zero, and $I_1^{H_E+1} \subseteq I_{H_E+1}$ proves nilpotence. Finally, ∇ preserves each support exponent. Thus it preserves J_E and induces a K -derivation on the quotient. $\square$

The qualifier “finite” refers to rank over $R = K[L]$. Polynomial blocks can have arbitrary degree, so Q_E need not be finite-dimensional over K . This is exactly the useful distinction for power-logarithmic computation: finitely many exponent blocks, each represented by a finite polynomial.

Corollary 11.3 (Finite residual certificate). *For a system (3), let $\bar{\Psi}$ be its induced system in Q_E . It has a unique solution in the image of I_1^d . Any candidate $\bar{U}$ in that ideal satisfying*

$$\bar{\Psi}(\bar{U}) = 0$$

has the exact target coefficients of U_ . Only input nonlinear degrees $|\nu| \leq H_E - 1$ can contribute when $H_E \geq 1$. The Newton iterate with $2^{n+1} - 1 > H_E$ automatically supplies such a candidate. For reversion, the Euler formula (24) truncates exactly after $n = H_E$ in this quotient.*

Proof. The image of the original solution is a solution. The Lipschitz estimate raises support depth by one, and after $H_E + 1$ increases the error is zero in Q_E ; hence a solution in the augmentation ideal is unique. A nonlinear term of degree $|\nu|$ has support depth at least $|\nu| + 1$, so terms with $|\nu| \geq H_E$ vanish. The Newton and Lagrange claims follow from their support-depth bounds. $\square$

This provides an independently checkable certificate: the checker need not reproduce the Newton trajectory. It can validate the finite dependence set, the complete relevant input coefficients, and a finite residual identity in Q_E . Such a certificate is only as effective as the supplied support-completeness proof; the next result makes that proof concrete for finite generators.

11.2 A rational certificate for finite supports

Theorem 11.4 (Finite supports admit a positive rational grading). *Suppose $S = \{s_1, \dots, s_k\}$ is finite and positive. The subgroup $H = \langle S \rangle \subseteq \Gamma$ is a finitely generated free abelian group. There exists an additive map*

$$w : H \longrightarrow \mathbb{Q}, \quad w(s_i) \geq 1 \quad (1 \leq i \leq k).$$

This map need not preserve the ambient order. For every $\alpha \in M(S)$,

$$H_S(\alpha) \leq \lfloor w(\alpha) \rfloor. \quad (38)$$

Given integer coordinates for the s_i and a rational vector representing w , checking the k inequalities supplies a finite certificate of this word-length bound.

Proof. Ordered abelian groups are torsion free, so their finitely generated subgroups are free abelian. Choose a basis of H and regard s_i as integer vectors in $\mathbb{R}^r$. Their convex hull does not contain zero. Indeed, a convex representation of zero by rational vectors would admit one with rational coefficients: reduce to an affinely independent subset of minimal size and solve its rational linear system. Clearing denominators would give a nontrivial relation $\sum_i n_i s_i = 0$ with all $n_i \geq 0$, impossible because every $s_i > 0$ in Γ .

A compact convex polytope not containing zero is strictly separated from zero by a real linear functional. One direct construction uses its nearest point p to zero: for every point x of the polytope, $p \cdot x \geq \|p\|^2 > 0$. Approximate the finitely many coefficients of this functional by rationals closely enough to preserve its strict positivity at the finitely many s_i . Then multiply by a positive rational constant so that all values are at least one. This gives w . For a word of length n summing to α , additivity gives $w(\alpha) \geq n$, proving (38). $\square$

Let the integer coordinate matrix have columns s_i . For a target $\alpha \in H$, all its generator-count representations occur among

$$m \in \mathbb{N}^k, \quad \sum_i m_i \leq \lfloor w(\alpha) \rfloor, \quad \sum_i m_i s_i = \alpha. \quad (39)$$

This is a finite search. It gives the exact depth by maximizing $\sum_i m_i$. The ordered words need not all be listed: for each count vector, its subcount vectors generate all corresponding divisor exponents. Their union over the target representations is exactly D_E .

11.3 A practical certificate format

For finite-support reversion, a support certificate can contain an integer basis for H , the generator columns, the rational row w , the target coordinates, and the complete count-vector lists in (39). An exact checker verifies the matrix identities, the inequalities $w(s_i) \geq 1$, and exhaustive enumeration within the finite budget. It then constructs D_E , its projected multiplication table, and the induced Euler operator from the character values and polynomial coefficient differentiation.

A solution certificate is a vector of polynomial blocks indexed by D_E . The checker verifies that its exponent-zero block is zero and its residual in the finite quotient is zero. Corollary 11.3

then certifies the requested coefficients. Alternatively, the checker may evaluate the finite Euler formula directly and compare it with a Newton output.

This design cleanly separates trusted finite arithmetic from support reasoning. It does not assert polynomial total running time. The number of exponent blocks, the number of support representations, and the bit sizes of coefficients can be large. The logarithmic claim concerns the number of Newton rounds in (9), not the cost of a round or the size of the final answer.

Example 11.5 (A resonant finite quotient). In lexicographic $\mathbb{Z}^2$, take

$$S = \{(0, 1), (1, -2), (1, 0)\}, \quad \alpha = (2, 1).$$

All generators are positive, although one second coordinate is negative. The functional $w(a, b) = 3a + b$ takes values 1, 1, 3 on them and gives $w(\alpha) = 7$. The three count vectors representing α are

$$(1, 0, 2), \quad (3, 1, 1), \quad (5, 2, 0).$$

Their lengths are 3, 5, 7, so $H_S(\alpha) = 7$. The monoid is $\{(a, b) : a \geq 0, b \geq -2a\}$. Its target-divisor set is

$$D_{\{\alpha\}} = \{(a, b) : a = 0, 1, 2, -2a \leq b \leq 5 - 2a\},$$

which has 18 elements. Thus the exact coefficient problem factors through an 18-dimensional free module over $K[L]$, even though the ambient valuation cutoff below $(2, 1)$ contains infinitely many powers of $(0, 1)$. Three Newton updates, reaching depth $15 > 7$, suffice uniformly in this quotient. The count-vector and quotient-rank assertions are checked by `verify.py` using exact integer arithmetic.

12 An infinite rank-two support with quadratic depth

The positive rational grading of Theorem 11.4 cannot in general be chosen on an infinite positive well-ordered support. A concrete family also shows how an exact nonlinear depth function replaces it.

Let $\Gamma = \mathbb{Z}\eta \oplus \mathbb{Z}\delta$ be lexicographically ordered, with $\eta = (1, 0)$ and $\delta = (0, 1)$. Define

$$S = \{\delta\} \cup \{s_n : n \geq 1\}, \quad s_n = n\eta - n^2\delta. \quad (40)$$

The set is positive and well ordered: δ is first, and the s_n are strictly increasing by their first coordinates. It has order type $1 + \omega$.

Theorem 12.1 (Exact monoid and nonlinear depth). *For the support (40),*

$$M(S) = \{k\delta : k \geq 0\} \cup \{n\eta + k\delta : n \geq 1, k \geq -n^2\}. \quad (41)$$

Its depth is

$$H_S(k\delta) = k \quad (k \geq 0), \quad \boxed{H_S(n\eta + k\delta) = n^2 + k + 1} \quad (n \geq 1, k \geq -n^2). \quad (42)$$

There is no additive map $w : \Gamma \rightarrow \mathbb{R}$ that is strictly positive on all of S .

Proof. Consider a word with a copies of δ and $r \geq 1$ remaining letters $s_{n_1}, \dots, s_{n_r}$. Its coordinates are

$$n = \sum_{j=1}^r n_j, \quad k = a - \sum_{j=1}^r n_j^2.$$

Since $\sum n_j^2 \leq n^2$, it has $k \geq -n^2$. Conversely, every point with $n \geq 1$ and $k \geq -n^2$ is represented by one s_n and $k + n^2$ copies of δ . Words with first coordinate zero consist only of δ 's. This proves (41).

For the displayed word, its length is

$$a + r = k + \sum_{j=1}^r n_j^2 + r.$$

For positive integers with sum n ,

$$n^2 - \sum_{j=1}^r n_j^2 = 2 \sum_{i < j} n_i n_j \geq r - 1.$$

For $r = 1$ the inequality is equality; for $r \geq 2$ it follows, for example, from $n_i n_j \geq 1$ and $r(r-1) \geq r-1$. Thus every representation length is at most $n^2 + k + 1$. The one-letter choice s_n followed by $k + n^2$ copies of δ attains the bound. For $n = 0$ the depth is plainly k . This proves (42).

Finally, if an additive w were positive on S , put $c = w(\delta) > 0$ and $d = w(\eta) \in \mathbb{R}$. Then

$$w(s_n) = nd - n^2 c < 0$$

for all sufficiently large n , a contradiction. $\square$

The obstruction is not simply that S is infinite, or that its exponent group has high rank. It is the impossibility of assigning finite real weights compatible with this entire infinite family. The depth function is still finite at every exponent and is explicitly computable.

Corollary 12.2 (A target-adaptive sharp rational certificate). *For the quadratic support, fix a target $n\eta + k\delta$ with $n \geq 1$. All its word representations use only $\{\delta, s_1, \dots, s_n\}$. The rational functional*

$$w_n(\delta) = 1, \quad w_n(\eta) = n + \frac{1}{n}$$

is at least one on these generators and gives the exact target depth:

$$w_n(n\eta + k\delta) = n^2 + k + 1.$$

Proof. First coordinates are nonnegative and add, so a word representing the chosen target cannot use s_j with $j > n$. For $1 \leq j \leq n$,

$$w_n(s_j) - 1 = (n - j) \left(j - \frac{1}{n} \right) \geq 0.$$

The asserted target value follows by substitution. It agrees with the attained depth in Theorem 12.1. $\square$

Thus a single positive real grading is impossible globally, while a sharp rational certificate exists for every finite target individually. The functional is allowed to depend on the target; suppressing that dependence would change the theorem.

12.1 Arbitrarily fast depth profiles in rank two

The quadratic example belongs to a larger family.

Theorem 12.3 (Superadditive depth-profile realization). *Let $(b_n)_{n \geq 1}$ be positive integers satisfying*

$$b_{m+n} \geq b_m + b_n \quad (m, n \geq 1).$$

In lexicographic $\mathbb{Z}\eta \oplus \mathbb{Z}\delta$, put

$$S_b = \{\delta\} \cup \{n\eta - (b_n - 1)\delta : n \geq 1\}.$$

This is a positive well-ordered support. Its monoid and depth are

$$\begin{aligned} M(S_b) &= \{k\delta : k \geq 0\} \cup \{n\eta + k\delta : n \geq 1, k \geq 1 - b_n\}, \\ H_{S_b}(k\delta) &= k, \\ H_{S_b}(n\eta + k\delta) &= k + b_n \quad (n \geq 1, k \geq 1 - b_n). \end{aligned}$$

A real-valued additive functional strictly positive on all of S_b exists if and only if $\sup_n b_n/n < \infty$.

Proof. Positivity and well ordering follow from the first coordinates exactly as for (40). A word with a copies of δ and $r \geq 1$ other letters of indices $n_1, \dots, n_r$ has

$$n = \sum_j n_j, \quad k = a - \sum_j (b_{n_j} - 1), \quad \text{length} = k + \sum_j b_{n_j}.$$

Repeated superadditivity gives $\sum_j b_{n_j} \leq b_n$. It also gives $k = a + r - \sum_j b_{n_j} \geq 1 - b_n$. Conversely, a single letter of index n and $k + b_n - 1$ copies of δ realize every permitted (n, k) and attain length $k + b_n$. Words with first coordinate zero contain only δ 's. This proves the monoid and depth formulas.

If a functional is positive on S_b , write its values on δ, η as $c > 0, d \in \mathbb{R}$. Positivity requires $b_n - 1 < nd/c$, so $b_n/n < d/c + 1/n$ is bounded above. Conversely, if $M = \sup_n b_n/n$ is finite, the functional with $w(\delta) = 1$ and $w(\eta) = M + 1$ satisfies

$$w(n\eta - (b_n - 1)\delta) = n(M + 1) - b_n + 1 \geq n + 1 > 0.$$

□

Corollary 12.4 (No rank-two growth ceiling). *Given any function $g : \mathbb{N}_{\geq 1} \rightarrow \mathbb{N}_{\geq 1}$, there is a positive well-ordered support in lexicographic $\mathbb{Z}^2$ such that*

$$H_S(n\eta) \geq g(n) \quad \text{for every } n \geq 1.$$

Proof. Set $c_n = \max_{1 \leq j \leq n} g(j)$ and $b_n = nc_n$. Because c_n is nondecreasing,

$$b_{m+n} = (m+n)c_{m+n} \geq mc_m + nc_n = b_m + b_n.$$

Theorem 12.3 gives depth $b_n \geq g(n)$ at $n\eta$.

□

For example, $b_n = n!$ is superadditive: for $m, n \geq 1$, $(m+n)! \geq 2m!n! \geq m! + n!$. It gives the exact depth $H_{S_b}(n\eta) = n!$. The quadratic support corresponds to $b_n = n^2 + 1$, which is also superadditive. These examples show that finite coefficient dependence does not imply a rank-dependent polynomial bound on the required nonlinear order.

Proposition 12.5 (The maximum depth can be noncancelling). *For any positive well-ordered S , put $\sigma = \sum_{s \in S} t^s$ over the coefficient field $\mathbb{Q}$. The small-coefficient equation*

$$V = \frac{\sigma}{1 - V}$$

has solution

$$V = \sum_{m \geq 1} \frac{1}{m} \binom{2m-2}{m-1} \sigma^m.$$

For every nonzero $\alpha \in M(S)$, the term with $m = H_S(\alpha)$ makes a strictly positive contribution to $[t^\alpha] V$.

Proof. The equation is equivalent to $V^2 - V + \sigma = 0$; its zero-constant branch gives the displayed Catalan series. Positive-word summability justifies specializing an ordinary formal variable to σ . The coefficient of t^α in σ^m counts ordered support words of length m with that sum. It is a nonnegative integer, finite by the word lemma, and is positive when $m = H_S(\alpha)$. The multiplying Catalan coefficient is positive. Thus the maximum-depth contribution is realized without cancellation. $\square$

12.2 A genuine infinite-support inverse

Choose $h = \delta$ and any character with $\chi(\delta) = 1$, for example $\chi(n\eta + k\delta) = n\omega + k$ over $K = \mathbb{Q}(\omega)$. Let

$$b = t^\delta + \sum_{n \geq 1} t^{n\eta - n^2\delta}, \quad B = Xb, \quad F = X + B. \quad (43)$$

Then $v(b) = \delta > 0$, so Theorem 8.1 applies. The inverse has normalized support in (41). For any $n \geq 1$ and $k \geq -n^2$, its coefficient at $n\eta + k\delta$ is determined by the Lagrange orders

$$1 \leq m \leq n^2 + k + 1,$$

and by Newton stage

$$j = \lfloor \log_2(n^2 + k + 2) \rfloor. \quad (44)$$

These are guaranteed support bounds, not assertions that every possible coefficient is nonzero or that cancellation cannot improve the round count for a particular coefficient.

There is also a direct effective enumeration. A target with first coordinate n can use only $s_1, \dots, s_n$. Enumerate integer partitions of n to determine their multiplicities; the required number of δ 's is then $a = k + \sum n_i^2$, which must be nonnegative. This finite calculation gives the exact depth and the relevant divisor set, without searching all support terms below a valuation cut. The verification program independently performs this partition enumeration for $1 \leq n \leq 10$ and $-n^2 \leq k \leq 10$.

Remark 12.6 (What this example rules out). One cannot replace the arbitrary-group theory by an argument that always first assigns a positive real additive weight to every input exponent. Such a weight exists for finite supports but not for (40). The word-depth proof and the finite target quotient remain valid, so this is an obstruction to a proposed proof and precision representation, not to inversion itself.

13 Exact verification and reproducibility

13.1 What was executed

The accompanying `verify.py` uses Python's exact rational arithmetic and polynomial arithmetic over $\mathbb{Q}[L]$ in SymPy. It performs no floating-point tests and makes no network requests. The recorded run used Python 3.13.5 and SymPy 1.14.0. It writes `verification_results.json`, including the tested ranges, coefficients, support lists, and assertions.

Four groups of checks were executed successfully. They concern the scalar sharpness example, a resonant power-logarithmic inverse, the finite quotient in Example 11.5, and the quadratic-depth formula in Theorems 12.1 and 12.3. These are finite audits of signs, indexing, normalization, resonance handling, and support enumeration. The quantified theorems depend on the proofs, not on extending a finite pattern by assumption.

13.2 Scalar sharpness through degree 130

The program represents ordinary power series through degree 130 by arrays of rational coefficients. It checks the exact root residual, then performs Newton updates for the rational equation $u + z/(1 + u) = 0$. The first nonzero errors are:

Newton index n	Error degree r_n	Leading coefficient	$(2r_n - 1)$
0	1	1	1
1	3	1	5
2	7	1	13
3	15	1	29
4	31	1	61
5	63	1	125
6	127	1	253

The last column is the multiple of h in the displacement-error valuation for (31). Every entry agrees with the proven formulas.

13.3 A resonant power–logarithmic example

For an independent block calculation, take lexicographic $\Gamma = \mathbb{Z}^2$, $h = (0, 1)$, and $\chi(a, b) = b$. Let

$$s_1 = (0, 2), \quad s_2 = (1, -3), \quad s_3 = s_1 + s_2 = (1, -1), \quad q_i = t^{s_i} \ (i = 1, 2),$$

and set

$$b = q_1(L + 1) + q_2(L^2 - 2) + q_1q_2L. \quad (45)$$

The third input block is a primitive coefficient at the same exponent as the product of the first two generators. It therefore tests a genuine support resonance, rather than a freely indexed three-variable model. The normalized feedback equation is

$$\begin{aligned} A_B(U) = & q_1(1 + U)^{-1}(L + \log(1 + U) + 1) \\ & + q_2(1 + U)^4((L + \log(1 + U))^2 - 2) \\ & + q_1q_2(1 + U)^2(L + \log(1 + U)). \end{aligned}$$

The program computes all 27 nonconstant blocks of total q_1, q_2 degree at most six. It compares the Euler–Lagrange formula, six Picard updates, and two Newton updates. All three results agree exactly. Both composition residuals vanish through the cutoff. In particular, it checks not only $F \circ G = X$ but also $G \circ F = X$, whose normalized residual is

$$b + (1 + b)T_B(U).$$

Some resulting blocks are:

Monomial	Coefficient in U
q_1	$-L - 1$
q_2	$2 - L^2$
$q_1 q_2$	$3L^3 + 6L^2 - 5L - 8$
q_1^2	$-L^2 - L$
q_2^2	$4L^4 + 2L^3 - 16L^2 - 4L + 16$
$q_1^2 q_2$	$-L^4 - 8L^3 - 13L^2 + 4L + 10$

No finite total-degree cutoff is claimed to represent the full valuation cut below an arbitrary lexicographic exponent. It is the finite quotient appropriate to this particular block audit.

13.4 Support and depth checks

For Example 11.5, exhaustive enumeration under the certified weight budget 7 gives exactly the three stated count vectors, maximum length 7, and 18 divisor exponents. A separate downward-closure check confirms that projected multiplication cannot discard a future contribution to a target.

For (40), the program enumerates the integer partitions of n and computes each feasible representation length independently of the closed formula. It checks 495 pairs with $1 \leq n \leq 10$ and $-n^2 \leq k \leq 10$. All maximum lengths equal $n^2 + k + 1$. A further exact partition audit tests the factorial profile $b_n = n!$ at 49 selected boundary and interior targets for $1 \leq n \leq 9$.

13.5 Rebuilding the package

From the package directory, run

```
python verify.py
pdflatex -interaction=nonstopmode -halt-on-error article.tex
pdflatex -interaction=nonstopmode -halt-on-error article.tex
pdflatex -interaction=nonstopmode -halt-on-error article.tex
```

The supplied build script performs the same steps. The TeX source contains its bibliography and tables; it does not require the repository checkout, external images, or a bibliography database. The verification output is a separate audit record rather than a hidden source of unstated hypotheses.

14 A statement-level formalization plan

The inspected Lean module `Analysis/Transseries/Lean/Transseries/TransseriesWellBased.lean` contains the binary support bridge: the named multiplicative declarations `neumann_isPW0` and `neumann_finite_factorizations`, their generated additive counterparts `neumann_add_isPW0` and `neumann_finite_decompositions`, Dickson-type declarations, and order-dual wrappers. Its namespace remains `Fabius` [3]. These are relevant foundations, not a formal proof of this article's inverse or Newton theorem.

A realistic extension proceeds through mathematical interfaces rather than marking an entire chapter as formalized because it mentions Hahn series.

Layer	Statements to formalize
All-word support	Higman embedding for finite words; positive-sum strictness; well-ordering of $M(S)$; finite all-length fibers.
Depth filtration	Maximum representation length; support ideals I_r ; separatedness; completeness and strong summation of rising-depth families.
Implicit systems	Summable evaluation; the one-factor and two-factor remainder estimates; unipotent matrix inverse; Newton depth $2r + 1$.
Differential blocks	Character data (h, χ) ; strong derivative; iterated block identity; induced support-preserving Euler derivation.
Substitution and reversion	Strong Taylor map; generator identities; associativity; justified parameter specialization; normalized Euler formula.
Finite certificates	Divisor closure; largest coefficient-annihilating ideal; finite quotient multiplication; nilpotence; residual checker.
Sharp boundaries	Exact scalar Newton error identity; noncofinal rank-two counterexample; uniform valuation criterion; quadratic-depth family.

The finite quotient theorem suggests a particularly manageable first implementation target. A checker for finite exponent lists, rational characters, polynomial blocks, and a supplied rational weight certificate can be separated from the more difficult abstract Hahn infrastructure. Its soundness theorem would explicitly invoke the completed support lemmas, rather than infer them from finite examples.

No new Lean module was compiled for this package. The exact Python checks are reproducible executable tests, and the mathematical arguments are conventional proofs. The distinction is important both for the current status report and for deciding which formalization dependencies remain.

15 Further research questions

The questions below are proposed continuations with explicit mathematical targets. Their appearance here does not assert that every formulation is an established open problem in the whole literature.

Research question 15.1 (Variable-shift derivations and genuine multi-scale calculus). Replace (12) by a strong derivation for which differentiating a monomial can produce several exponent shifts depending on that monomial. Which well-founded condition on the resulting support-dependency graph replaces the common positive monoid? Seek a substitution theorem with finite anchored-dependency fibers and a computable analogue of H_S , without requiring all shifts to lie in one Archimedean class.

The current proof uses a fixed shift to cancel the $-h$ normalization in every Taylor order. A variable-shift extension needs a replacement for that cancellation, not merely a change of notation.

Research question 15.2 (When does ordinary sequential iteration suffice?). For the small-coefficient systems here, an ordinary sequence indexed by $\mathbb{N}$ reaches every coefficient; transfinite Newton stages are unnecessary. Characterize strong nonlinear operators outside this class for which the same statement remains true. Can one isolate an intrinsic finite coefficient-dependency condition equivalent to stabilization after ordinary sequential iteration, and distinguish it from the existence of an abstract fixed point obtained by a different construction?

Research question 15.3 (Cancellation-sensitive precision). The depth H_S is a support-only invariant and ignores coefficients. Find efficiently checkable hypotheses under which the coefficient at a maximum-depth target is nonzero, and a certificate format for cancellations that reduce the necessary Newton depth. The scalar sharpness theorem shows that no unconditional improvement is possible, but structured families may admit much smaller certified dependence sets.

Research question 15.4 (Effective infinite supports). For which computable descriptions of infinite well-ordered supports can one decide membership in $M(S)$, compute $H_S(\alpha)$, and enumerate $D_S(E)$? Presburger descriptions, finite-state descriptions, and polynomially parameterized lexicographic supports are natural test classes. The quadratic family gives a nontrivial solvable example; existential finiteness alone is not an algorithm.

Research question 15.5 (Compressed infinite cuts). When Theorem 10.1 rules out a finite monomial list, determine when a cutoff nevertheless admits a finite algebraic, rational, or differential block representation that is stable under substitution. A useful objective is a certificate calculus whose objects combine finite divisor quotients with exact infinite lower-rank blocks, so that crossing a noncofinal cut does not force term-by-term enumeration.

Research question 15.6 (Bit complexity of minimal coefficient quotients). Give output-sensitive complexity bounds for constructing D_E , multiplying in Q_E , and checking residual certificates. Determine when resonance compression saves work compared with a free multi-index expansion. The logarithmic number of Newton rounds is established here; polynomial total complexity is not.

Research question 15.7 (Analytic realization and differential characters). Identify coefficient fields, characters, support classes, and evaluation maps for which the formal substitution and inverse agree with operations on actual Hardy-field or sectorial analytic germs. In a non-Archimedean support hierarchy, an injective character into an abstract coefficient field need not be an analytic exponent assignment. A successful realization theorem must address both the ordered monomial hierarchy and the coefficient field, not only the formal support estimates.

Research question 15.8 (Divided powers and positive characteristic). The word-support and finite-quotient arguments do not intrinsically require factorials. Can the reversion theorem be expressed using Hasse–Schmidt operators or divided-power structures so as to isolate exactly what survives in positive characteristic? In particular, distinguish the support-theoretic implicit theorem from the characteristic-zero Euler and Lagrange formulas.

Research question 15.9 (An independently checked finite certificate engine). Implement the quotient certificate of Section 11 in a proof assistant, beginning with integer-coordinate exponents, rational characters, and polynomial logarithmic blocks. The first milestone should be a kernel-checked implication from a complete finite support certificate and a zero residual to exact target coefficients, including resonances. A subsequent milestone is certificate synthesis rather than merely certificate checking.

16 Conclusion

The repository’s non-Archimedean reversion question has a positive answer in the stated fixed-shift Hahn power–logarithmic setting: cofinal valuation improvement is not needed for existence, the ordinary integer Lagrange index is sufficient, and the inverse remains inside the normalized input support monoid. The correct coefficientwise finiteness invariant is maximum positive-word length.

That invariant produces more than an existence proof. It yields the sharp Newton law $r \mapsto 2r + 1$, an exact uniform criterion for valuation convergence, and a minimal finite quotient for

any finite coefficient request. The quadratic-depth example demonstrates that no global positive real additive grading can always replace this combinatorics, even in a rank-two group. The superadditive-profile construction allows arbitrary depth growth in that same rank. The result is a support and precision calculus that separates valid formal computation from stronger topological or analytic claims and supplies a concrete next target for verified symbolic implementation.

A The coefficient Lagrange identity over a general $\mathbb{Q}$ -algebra

For clarity, we record a proof that does not assume the constant coefficient of the relevant series is invertible after specialization.

Lemma A.1 (Formal coefficient inversion). *Let A be any commutative $\mathbb{Q}$ -algebra and $\phi(y) \in A[[y]]$. There is a unique $c(z) \in zA[[z]]$ satisfying $c = z\phi(c)$, and*

$$[z^n]c(z) = \frac{1}{n}[y^{n-1}]\phi(y)^n \quad (n \geq 1).$$

Proof. Successive comparison of coefficients gives existence and uniqueness: the coefficient at z^n on the right depends only on coefficients of c of degrees less than n . Moreover, each coefficient is a polynomial with rational coefficients in finitely many coefficients of ϕ .

First work in the universal polynomial ring $A_0 = \mathbb{Q}[a_0, a_1, a_2, \dots]$, with $\phi(y) = \sum_{j \geq 0} a_j y^j$, and localize by inverting a_0 . Then $z = y/\phi(y)$ is a formal change of parameter with invertible linear coefficient. Formal residues obey change of variable:

$$\text{Res}_z f(z) dz = \text{Res}_y f(z(y)) z'(y) dy.$$

For Laurent monomials other than z^{-1} this follows because the substituted expression is a derivative and has zero residue; for z^{-1} the logarithmic derivative has residue one. Linearity gives the identity for a Laurent series with finite principal part, as used below.

Using $c(z(y)) = y$, we obtain

$$\begin{aligned} [z^n]c(z) &= \text{Res}_z c(z) z^{-n-1} dz \\ &= \text{Res}_y y^{-n} (\phi(y)^n - y\phi(y)^{n-1}\phi'(y)) dy \\ &= [y^{n-1}]\phi(y)^n - [y^{n-2}]\phi(y)^{n-1}\phi'(y) \\ &= \frac{1}{n}[y^{n-1}]\phi(y)^n. \end{aligned}$$

The last equality follows by differentiating ϕ^n ; it also holds at $n = 1$ with the negative-index coefficient interpreted as zero. Both sides of the resulting coefficient identity lie in A_0 , and A_0 injects into its localization. Hence the identity already holds in A_0 . Specialize its finitely many relevant indeterminates to any coefficients in A to obtain the asserted formula without any invertibility condition on $\phi(0)$. Replacing ϕ by $-\phi$ supplies the sign $(-1)^n$ used in (23). $\square$

B Source and verification status ledger

Item	Status and boundary
Repository snapshot	Pinned to commit <code>3f2d57344dc6f0b3dfa376f9490bfe13d1beb81e</code> . The recent reversion question, README, and support Lean module were inspected; no exhaustive repository audit is claimed.
Classical support background	Higman word embeddings, Hahn finite-fiber summation, and ordinary coefficient Lagrange inversion. Proofs of the specific versions used here are included.
Conventional proofs	Theorems on support depth, implicit systems, fixed-shift substitution and reversion, Newton precision, valuation and cut criteria, minimal finite quotients, and the quadratic-depth family.
Executed exact checks	Scalar Newton through ordinary degree 130; resonant block inversion through total degree 6; the 18-block finite quotient; 495 quadratic-profile and 49 factorial-profile depth comparisons.
Not executed or not claimed	No new Lean compilation, no universal analytic realization, no Borel or resurgence theorem, no complete literature-priority determination, and no polynomial-time complexity claim.
Reproducible materials	Standalone <code>article.tex</code> , compiled PDF, <code>verify.py</code> , exact JSON results, build script, requirements, README, and provenance record.

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